

Let's talk about the weather

X19 (2019) — English translation

The map below shows isobars (for the same altitude, approximately at sea level). Assume that the isobar diagram is stationary (changing slowly over time).

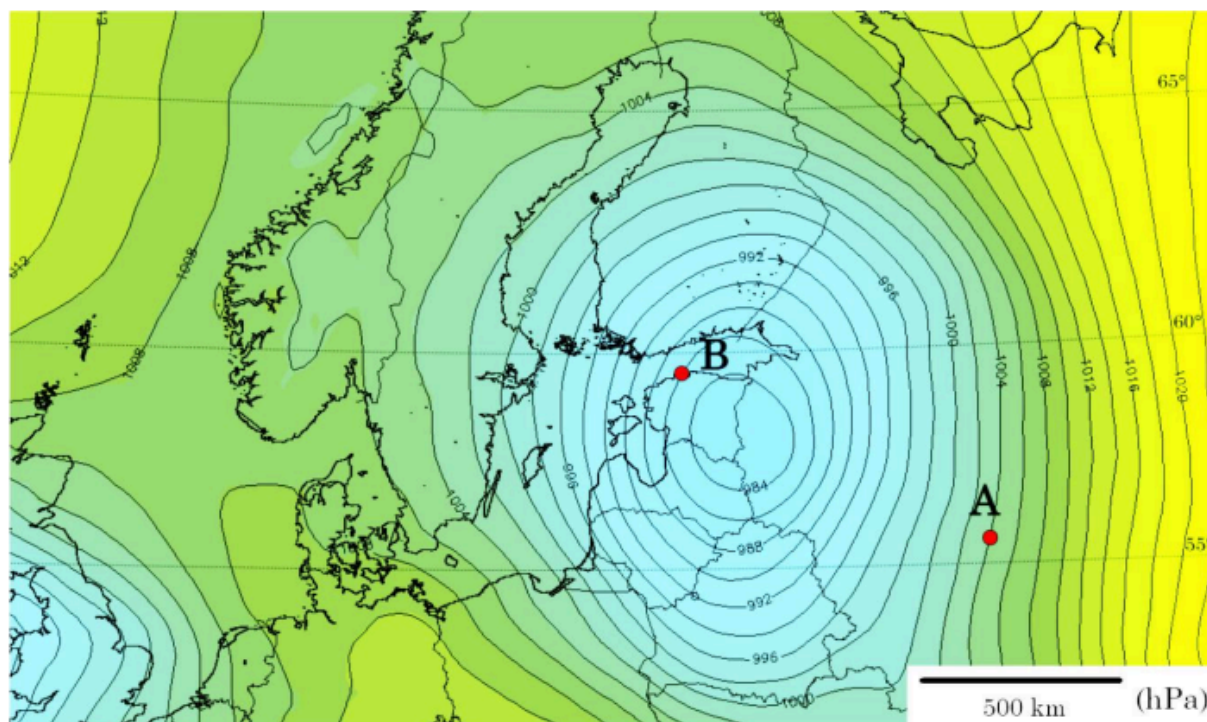


Fig. 1.

A1	Sketch the wind direction at points <i>A</i> and <i>B</i> .	1.00
A2	Estimate the wind speed v_A at point <i>A</i> . Consider the isobars to be straight lines around this point. Air density $\rho = 1.23 \text{ kg/m}^3$.	2.00
A3	Estimate the wind speed v_B at point <i>B</i> .	3.00

Note: When a body of mass m (for example, a layer of air) moves with speed v in a rotating frame of reference (rotating with angular speed Ω), it is acted upon by a certain fictitious force, which is called the Coriolis force:

$$F_C = 2m\Omega v \sin \varphi$$

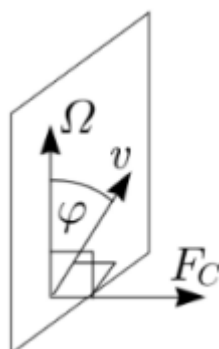


Fig. 2.

Source: <https://pho.rs/p/3056>